

Name: _____

Date: _____

Class: _____

Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	Prototype must be easy to maintain	1. Teach someone how to maintain the prototype 2. Time the amount of time it takes for the person to maintain the prototype 3. Check the results of the maintenance	a. Prototype is very dirty b. Prototype is broken c. Prototype is very clean	Prototype is cleaned, dirty, or very dirty. Prototype should be intact.	Pass: Prototype is clean and functional Fail: Prototype is dirty Goal: Prototype is clean and functional
2	Prototype must water the plants well	1. Place watering system over a large bucket 2. Turn the watering system on for an hour and measure how much water is in the bucket		a. Bucket has 1-2 inches of water b. Bucket has more than 2 inches of water c. Bucket has less than an inch of water	Pass: 1-2 inches of water Fail: Less than an inch of water or more than 2 inches of water Goal: Bucket has 1-2 inches of water in it
3	Prototype must be robust	1. Turn the watering system on 2. Turn the watering system off	a. 1 repetition b. 2 repetitions	a. Watering system is intact b. Watering system is not intact	Pass: Watering system is not broken Fail: Watering system is broken Goal: Watering system is intact
4	Prototype must water a wide	1. Place sprinklers on prototype 2. Place prototype above 10	a. 1 repetition	a. All are evenly wet	Pass: Most papers are wet

	area	sheets of paper spread out to cover a square area 3. Turn prototype on 4. See if the papers are evenly wet		b. All are not evenly wet	Fail: Most papers are not wet Goal: All papers are wet
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Name of your field expert:

Xu Song

Why are they a field expert?

Experience in engineering and gardening.

What did they say to you about the tests you are proposing?

They said I could add more test cases for Test 2.

1	Prototype must be easy to maintain	4. Teach someone how to maintain the prototype 5. Time the amount of time it takes for the person to maintain the prototype 6. Check the results of the maintenance	d. Prototype is very dirty e. Prototype is broken f. Prototype is very clean	Prototype is cleaned, dirty, or very dirty. Prototype should be intact.	Pass: Prototype is clean and functional Fail: Prototype is dirty Goal: Prototype is clean and functional
2	Prototype must water the plants well	3. Place watering system over a large bucket 4. Turn the watering system on for an hour and measure how much water is in the bucket	a. 1 bucket b. 3 buckets side by side	d. Bucket has 1-2 inches of water e. Bucket has more than 2 inches of water f. Bucket has less than an inch of water	Pass: 1-2 inches of water Fail: Less than an inch of water or more than 2 inches of water Goal: Bucket has 1-2 inches of water in it
3	Prototype must be robust	3. Turn the watering system on 4. Turn the watering system off	c. 1 repetition d. 2	c. Watering system is intact d. Watering	Pass: Watering system is not broken Fail: Watering

			repetitions	system is not intact	system is broken Goal: Watering system is intact
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	5	4	3	2	1	0
1. Addressing Requirements. The final prototype Testing Plan addresses _____ of the <i>high priority</i> design requirements.	<i>all or nearly all</i>	<i>many</i>	<i>some</i>	<i>a few</i>	<i>one</i>	<i>none - or is missing altogether</i>
2. Conduct of Tests. The Testing Plan _____ in describing the conduct of tests (through physical	<i>is effective</i>	<i>is generally effective</i>	<i>is only adequate</i>	<i>is partially effective</i>	<i>is at least minimally effective</i>	<i>is void</i>

and/or mathematical modeling).						
3. Quantity of Tests Planned. The Testing Plan defines _____ of <i>all</i> design requirements based on the instructional context.	sufficient quantity to feasibly test 100%	sufficient quantity to feasibly test 80%	sufficient quantity to feasibly test 60%	sufficient quantity to feasibly test 40%	<i>sufficient quantity to feasibly test 20%</i>	<i>no feasible test</i>
4. Logic of the Plan. The Testing Plan provides _____.	a logical and well-developed explanation	<i>a logical and generally developed explanation</i>	<i>a generally logical and adequately developed explanation</i>	<i>a somewhat logical and partially-developed explanation</i>	<i>at least a generally logical and/or partially-developed explanation</i>	<i>no explanation</i>
5. Use of Field Experts. The Testing Plan is confirmed by _____.	<i>more than two</i> field experts with feedback provided and incorporated	<i>two</i> field experts with feedback provided and incorporated	one field expert with feedback provided and incorporated	one field expert with feedback provided	<i>one field expert but no response was documented</i>	<i>Field expertise is not mentioned or addressed and field experts are not consulted.</i>